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# Enhancing Wear Performance of W-Cu Composites through Response Surface Methodology

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***Abstract**:* This paper aims to maximize wear resistance of tungsten-copper (W-Cu) composites on Response Surface Methodology (RSM) and Box-Behnken Design (BBD). The experimental variables included the reinforcement percentage (2040), the temperature (160-20024°C) and the mechanical load (80100 N). Regression and ANOVA analysis showed that temperature ($p = 0.001$) and percentage of Cu W ($p = 0.038$) are significant variables that influence the wear rate, whereas quadratic terms showed weaker influences ($p > 0.2$). The optimal conditions were established as 40% Cu, 200 0 C and 100 N load with the minimum wear rate of 3.364 mm 3 /min and a desirability of 1.0. The synergistic effects of temperature and reinforcement content on wear behavior were demonstrated with the help of contour and surface plots. It is found that RSM is efficient and minimizes experimental trials with accurate optimization to allow the production of W-Cu composites with an excellent level of durability in electrical contacts and high-temperature tooling.

***Keywords:*** *Tungsten–copper composites, Wear rate optimization, Response Surface Methodology, Box–Behnken Design, Design of Experiments.*

## 1. INTRODUCTION

The components of tungsten 2 copper (W CU) alloys serve as areas where metallic components are used in most contemporary engineering solutions because of the complementary nature of the constituent components. Copper matrix consists of material that is excellent in electrical and thermal conductivity, tungsten particles give high level of strength, thermal stability and wear resistance. These composites play an important role in electrical contacts, heat sinks and high-temperature parts where conductivity and life are needed to be balanced [2].

Another important application of wear resistance is in sliding electrical contacts as well as other tribological applications. As it has been in the past studies, the wear performance is a factor of the tungsten content, the temperature of the service and the load [3]. These parameters tend to be interdependent and that is why the classical methods of experimentation (OFAT) cannot be used to model the complex, nonlinear interactions [4].

Recent developments focus on systematic design as Design of Experiments (DOE) and Response Surface Methodology (RSM). In contrast to OFAT, RSM is able to capture second-order interactions as well as nonlinear interactions between composition and service conditions [5,6]. In this work, a BoxBehnken Design (BBD) within the framework of RSM was used to predict and optimize the wear behavior of WCu composites [7].

The present work investigates:

1. The interaction and the individual effects of CuW content, temperature and load on wear resistance.
2. Wear rate prediction at different conditions.
3. Determining the best composition and service parameters to achieve better lifetime.

The general aim is to ensure the formation of structure-property relationship of W-Cu composites.

## 2. METHODS AND METHODOLOGY

### 2.1 Optimization Technique

The methods of optimization are important in developing new products, designing processes, and enhancing performance. To achieve wear resistance, methods like Build Test Fix (BTF), One Factor at a Time (OFAT) and Design of Experiments (DOE) are popular [813]. Nonetheless, DOE, especially RSM, is more efficient because it can consider numerous parameters at the same time and the number of experimental runs is minimized.

RSM with Box-Behnken Design was used in this experiment to examine three variables (reinforcement, temperature, load) and their interaction to obtain the optimal wear resistance.

### 2.2 Materials

The CuW composites were made through the powder metallurgy since it could make the combination of the exceptional properties of the two metals. Tungsten and copper offer thermal and mechanical strength and electrical and thermal conductivity, respectively [14].

The process involves:

1. Compaction: Tungsten powders are forced into the required shape.
2. Sintering: The high-temperature sintering of the compacted tungsten is done to form a porous structure by bonding the particles.
3. Infiltration: The molten copper is infiltrated in to the porous tungsten skeleton and then solidified to form a CuW composite system that has a uniform distribution [15].

This technique guarantees a foamed CuW architecture that has a high phase bonding and homogenous performance.

### 2.3 Experimental Setup

All the experiments were conducted on tribometer setup


***Figure 1-Tribometer Setup***

## 3. RESULTS AND DISCUSSION

Minitab statistical software was used to formulate spreadsheet predictive models on wear rate. ANOVA (Analysis of Variance) was used to determine the significance of each parameter and a linear regression model was developed to predict the output response [16]. The statistical method facilitated proper assessment of the impact of the reinforcement percentage, temperature, and load to the wear resistance of W-Cu composite.

### 3.1 W-Cu Composite Wear Optimization Desirability Values Analysis.

The desirability analysis made some valuable contributions to the enhancement of W-Cu composites in regard to wear resistance. The resulting outcomes showed that the maximum desirability was achieved in the specimen with 30 percent tungsten and under the conditions of 200 °C and 100 N weight. This implies that the conditions are the most desirable set of properties.

The composites of 40% tungsten showed moderate values of desirability of between 0.468-0.817, and the best performance was at elevated temperatures and loads.

The desirability values were found to be low because of poor reinforcement and low load-bearing capacity of 20 percent tungsten composites with the range of 0.08 to 0.603.

In general, the temperature was found to have the most considerable effect, with all compositions being significantly better at 200 o C than they were at 160 °C. An increase in tungsten level contributed to the increase in wear resistance but the most consistent and effective performance was obtained with 30% tungsten in different thermal and mechanical conditions.

Such results correspond to the wear mechanisms of high-desirability conditions that were reported to have a higher proportion of mixed abrasive-oxidative wear over adhesive wear [17]. This wear change at optimum conditions is yet another reaffirmation of the usefulness of the proposed composite design.

***Table1-L9 Signal to Noise Ratio of wear rate***

| Experiments | Inputs Factors | | | Output Factors | |
|---|---|---|---|---|---|
| Trial No. | Reinforcement (%) | Temperature (°C) | Load (N) | Wear Rate (Nm/mm$^3$) | DESIR |
| 1 | 40 | 180 | 80 | 4.022 | 0.468400 |
| 2 | 40 | 200 | 90 | 3.657 | 0.767491 |
| 3 | 20 | 160 | 90 | 4.472 | 0.079991 |
| 4 | 30 | 200 | 80 | 3.801 | 0.624417 |
| 5 | 20 | 180 | 80 | 4.197 | 0.254548 |
| 6 | 20 | 180 | 100 | 3.719 | 0.603428 |
| 7 | 30 | 160 | 80 | 4.262 | 0.150770 |
| 8 | 30 | 200 | 100 | 3.400 | 0.973298 |
| 9 | 20 | 200 | 90 | 3.901 | 0.553638 |
| 10 | 30 | 160 | 100 | 4.015 | 0.499650 |
| 11 | 30 | 180 | 90 | 4.163 | 0.490050 |
| 12 | 30 | 180 | 90 | 3.743 | 0.490050 |
| 13 | 40 | 160 | 90 | 4.041 | 0.293843 |
| 14 | 40 | 180 | 100 | 3.652 | 0.817281 |
| 15 | 30 | 180 | 90 | 3.934 | 0.490050 |

### 3.2 Main Effects of Wear Rate

The plot of the main effects (Figure 2) shows the effect of each parameter separately on the wear rate of W -Cu composites. The parameter values were found to be the lowest wear rate at 200 o C (level 3), 100 N (level 3), and 40 percent reinforcement (level 3).

The regression model, designed to be used in the analysis, is presented below:

The plot indicates that:

1. Reinforcement content (20 -40%): As tungsten reinforcement is increased, the mean wear rate is reduced progressively. This validates the fact that tungsten is a strong enhancer of a wear resistance which is brought about by a hardening effect.
2. Temperature (160 200 o C): Wear rates increased at 160 o C, and decreased at 200 o C, which was due to the formation of protective layers of oxide and softening of the copper matrix at higher temperatures.
3. Load (80 100 N): With increasing load, the wear rate increased gradually because with greater mechanical stress, wear damage, although enhanced by tungsten reinforcement, is valued.

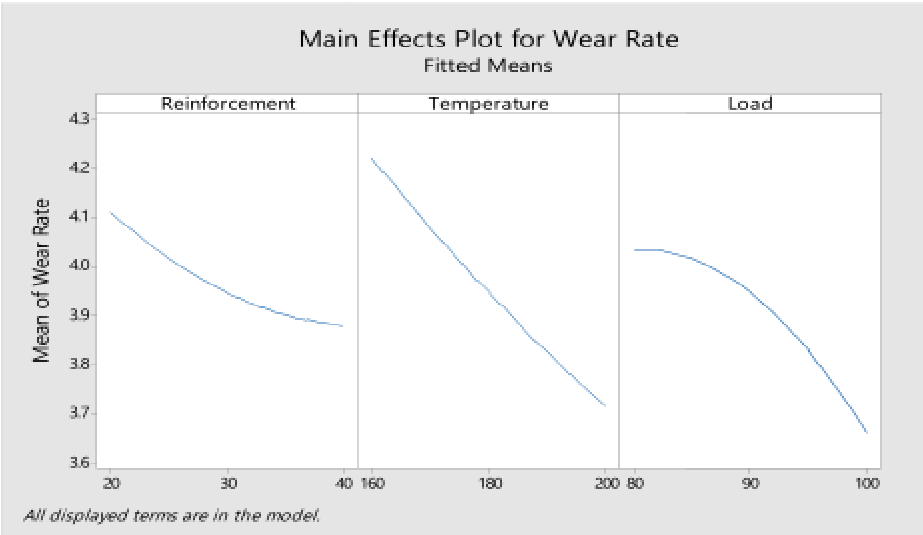


***Figure 2 Main Effects Plot of Wear rate.***

The trends validate the previous optimization findings with tungsten percentage and temperature prevailing as the key determinants of wear rate with load being secondary but significant. The wear resistance was maximum in the case of high tungsten concentration (40%), high temperature (200 o C), and moderate load (100 N). The trends are urgent in determining appropriate material composition and working conditions in the practical applications of W -Cu composites.

### 3.3 Analysis of Variance (ANOVA)

The ANOVA results (Table 2) validated the significance of the developed regression model. The model was found statistically significant at p = 0.003, confirming its suitability for predicting wear behavior.

***Table 2 – ANOVA Results for Wear Rate***

| Source | DF | Adj SS | Adj MS | F-Value | P-Value |
|---|---|---|---|---|---|
| Model | 6 | 0.95093 | 0.158488 | 9.30 | 0.003 |
| Linear | 3 | 0.90048 | 0.300161 | 17.62 | 0.001 |
| Reinforcement | 1 | 0.10511 | 0.105111 | 6.17 | 0.038 |
| Temperature | 1 | 0.51562 | 0.515620 | 30.27 | 0.001 |
| Load | 1 | 0.27975 | 0.279752 | 16.42 | 0.004 |
| Square | 3 | 0.05044 | 0.016815 | 0.99 | 0.446 |
| Reinforcement*Reinforcement | 1 | 0.00906 | 0.009062 | 0.53 | 0.487 |
| Temperature*Temperature | 1 | 0.00171 | 0.001713 | 0.10 | 0.759 |
| Load*Load | 1 | 0.03598 | 0.035975 | 2.11 | 0.184 |
| Error | 8 | 0.13629 | 0.017037 | – | – |
| Lack-of-Fit | 6 | 0.04785 | 0.007976 | 0.18 | 0.957 |
| Pure Error | 2 | 0.08844 | 0.044220 | – | – |
| Total | 14 | 1.08722 | – | – | – |

1. Temperature had the strongest effect on wear rate (F = 30.27, p = 0.001).

2. Load (F = 16.42, p = 0.004) and reinforcement percentage (F = 6.17, p = 0.038) were also significant contributors.
3. Quadratic terms were not statistically significant (p > 0.446), implying that the factor–response relationships are predominantly linear rather than curvilinear.
4. The lack-of-fit test (p = 0.957) indicated that the developed model fits well with the experimental data.

Thus, the ANOVA confirms that temperature is the most sensitive factor, followed by load and reinforcement content, in influencing the wear resistance of W–Cu composites [18].

### 3.4 Contour Plot for Wear Rate

Analyzing the contour plots, it is possible to obtain the further knowledge about the interaction of the factors on the wear rate of the W-Cu composite [19–23].

- Out of the Temperature x Reinforcement interaction, the minimum wear rate (< 3.50) was realised at tungsten content 3540 percent with high-temperature (190-200°C). This corroborates the hypothesis that reinforcement and level of tungsten increase in conjunction with an increase in temperature of operation will increase the wear resistance.
- The Load x Reinforcement plot shows that there is a positive effect of tungsten content. The relatively low wear rates sustained at 40% reinforcement ranging between 3.50–3.75 were maintained even at a high load of 100 N. Conversely, the lower reinforcement rates resulted in much higher wear rates at equivalent loads demonstrating the importance of having tungsten in strengthening the composite.
- According to the Load × Temperature plot, the wear rates cannot be less than 3.75 in case of the application of temperatures 190 o C and the moderate loads (8090 N). This means that increased levels of temperature trigger protective processes like development of the oxide layer that offsets the impact of increased load.

All these contour plots indicate that optimization of wear resistance is achievable through the choice of appropriate combination of process parameters:

1. Tungsten reinforced 35% or higher (hardness and ability to carry a load),
2. Temperature ≥ 190°C (to activate protective oxide mechanisms), and
3. Load < 90 N (to reduce wear caused by mechanical stress).


*Figure 3. Contour Plots for Wear Rate.*

The non-parallelism of contour lines is also a confirmation that there are important effects of interaction between factors. The most salient interaction is between temperature and reinforcement content meaning that optimization cannot be done at the parameter level, but must be done considering several parameters at the same time.

### 3.5 Development of Regression Model

Minitab software was employed to develop a regression model for predicting wear rate. By substituting experimental values of the parameters into the regression equation, wear rates at all levels of study parameters can be estimated. The correlation between predicted and experimental wear rates was analyzed, demonstrating the validity of the model.

The regression equation for wear rate is as follows:

$$Wear\ Rate = 2.45 - 0.0412 Reinforcement - 0.0321 Temperature + 0.159 Load + 0.000495\ Reinforcement * Reinforcement + 0.000054 Temperature * Temperature - 0.000987 Load * Load$$

***Table 3 – Experimental and Predicted Wear Rate***

| Set | Wear Rate (Experimental) | Wear Rate (Predicted) |
|---|---|---|
| 1 | 4.022 | 3.968 |
| 2 | 3.657 | 3.649 |
| 3 | 4.472 | 4.385 |
| 4 | 3.801 | 3.802 |
| 5 | 4.197 | 4.198 |
| 6 | 3.719 | 3.825 |
| 7 | 4.262 | 4.309 |
| 8 | 3.400 | 3.429 |
| 9 | 3.901 | 3.879 |
| 10 | 4.015 | 3.935 |
| 11 | 4.163 | 3.946 |
| 12 | 3.743 | 3.946 |
| 13 | 4.041 | 4.155 |
| 14 | 3.652 | 3.595 |
| 15 | 3.934 | 3.946 |

### 3.7 Confirmation Experiment Result

***Table 5 Confirmation Experiment Result***

| Parameter | Experimental Value | Predicted Value | Error (%) |
|---|---|---|---|
| Wear Rate | 3.416 | 3.363 | 1.57 |

Confirmation experiment was conducted by maintaining the parameters at the optimal levels suggested by the optimization method. The experimental wear rate (3.416) closely matched the predicted value (3.363), with an error of only 1.57%, validating the accuracy and reliability of the regression model.

## 4. CONCLUSIONS

This paper shows one of the potential ways of improving wear resistance of W-Cu composites by using Response Surface Methodology (RSM) and Box-Behnken Design. The main findings are summed up as below:

1. The best wear resistance parameters were obtained at the point of 40% reinforcement, 200°C temperatures and 100N load where the wear rate was 3.3635 and the desirability value was 1.0.

2. Effects of parameters: The statistical analysis revealed that the most important influence is temperature on the wear resistance then the reinforcement percentage and then load.
3. Mechanism Insights: The increased tungsten enhances wear resistance by hardening, and increased temperature supports the formation of oxide layer and softening W-Cu matrix, which minimized wear.
4. Regression Model Validity: The linear regression model built is a good predictor of wear rate with a difference value of less than 10 percent between the model prediction and the experimental values.
5. Optimization Findings: The highest wear performance is achieved at the limits of the experimented parameters, and this proves the impact of the combination of reinforcement, temperature and load.
6. The findings are useful to provide practical advice on the design of W-Cu composites to be used in high-temperature wear-critical environments, including electrical contacts and tools.

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